

Module 18: Resonance Ladder Certificate

Sweep $\beta = 299 + k \cdot \pi/10$ for k in $[0.50, 3.50]$ | May 2026

Battle Plan v1.6 -- David Fox

For each k , compute $S_\beta = \{p \text{ prime} \leq 100000 : \|p\beta\| < 1/p\}$, the BC sum $C(\beta) = \sum \log(p) \cdot p/(p-1)$ for p in S_β , and the RH genus bound $g_{\max} = \lceil C^2/4 \rceil$. Float64 sweep; key rows certified at 100 dps with mpmath 1.3.0. Cross-verified against M5 (SHA 9df98a39...) and M9 (SHA 5e39f3a9...).

Annotation Correction: k_c and $c/10^6$

An external conversation annotated $k=2.67$ as ' $\beta \sim c/10^6$ '. This is incorrect. The certified value is:

Quantity	Value	Note
$c / 10^6$	299.792458	exact, SI definition
$\beta_0 = 299 + \pi/10$	299.314159...	$k=1.00$
$k_c = (c/10^6 - 299)/(\pi/10)$	2.522472...	mpmath certified
$k=2.67$ gives β	299.838805	0.046 above $c/10^6$ = DIFFERENT
$k=2.67$ annotation	WRONG	$k_c = 2.52$, not 2.67

Certified Key Rows (mpmath 100 dps)

Full sweep: primes ≤ 100000 ; BC sum $C(\beta) = \sum \log(p) \cdot p/(p-1)$; $g_{\max} = \lceil C^2/4 \rceil$ from BC bound [consistent with M9 $g=33$ at $k=1.00$].

k	β	$ S_\beta $	$C(\beta)$	$g_{\max}$	Note
1.00	299.314159	4	11.42214869	33	β_0 [M5/M9 CERTIFIED]
1.25	299.392699	4	8.324027	18	local low
2.00	299.628319	6	16.30404412	67	matches ext. screenshot
2.52	299.791681	6	33.59099989	283	k_c : $\beta \sim c/10^6$ CORRECT
2.67	299.838805	6	28.32425777	201	ext. screenshot: wrong (see below)
2.72	299.854513	2	3.656523	4	local minimum (2 primes)
3.00	299.942478	4	11.01473296	31	matches ext. screenshot
3.13	299.983319	5	11.464079	33	local match
3.18	299.999026	14	58.25508311	849	explosion (ext: 11/29.17/372)
3.50	300.099557	2	3.034213	3	beyond 300

Green row = M5/M9 cross-check PASS. Blue row = certified k_c . Red row = explosion at $k=3.18$.

Discrepancy with External Screenshot Values

An external AI chat reported different values at $k=2.67$ and $k=3.18$. This is explained by the prime bound used:

k	Quantity	External AI	Our Certified	Explanation
2.67	$ S_\beta $	4	6	Ext. AI: primes $\sim \leq 191$ only
2.67	$C(\beta)$	9.217	28.32	Ext. $S=\{2,5,7,31\}$; we find 2 more primes
2.67	$g_{\max}$	37.2	201	$g \sim C^2/4$; higher $C \Rightarrow$ higher g
3.18	$ S_\beta $	11	14	3 extra primes from full search
3.18	$C(\beta)$	29.165	58.26	Large primes have big C_p contribution
3.18	$g_{\max}$	372.4	849	Consistent with corrected C
k_c	k value	2.67	2.52	$c/10^6=299.792458 \Rightarrow k=(0.792458)/(\pi/10)$

Note: At $k=2.67$, $C_{\text{ext}} = 9.217$ matches exactly $C_2+C_5+C_7+C_{31} = 1.386+2.012+2.270+3.547 = 9.215$ (round-off matches 9.217). The external AI's S_β at $k=2.67 = \{2,5,7,31\}$, missing primes > 191 . The VALUES AT $k=2.00$ AND $k=3.00$ AGREE, confirming shared methodology for primes covered by both searches.

What the Ladder Reveals (Certified)

Finding	Description	Certified
1. k=1.00 exact match	C=11.422, g_max=33, S={2,3,19,191}	M5 SHA 9df98a39, M9 SHA 5e39f3a9
2. Explosion near beta=300	k=3.18: S =14, C=58.26, g_max=849	mpmath certified, m18.out
3. k_c correct value	c/10^6 at k=2.5225, not k=2.67	mpmath: (299.792458-299)/(pi/10)
4. S fluctuates sharply	jumps between 2 and 14 in [k=0.5, 3.5]	float64 sweep, mpmath spot-checks
5. Near-300 regime	k in [3.0, 3.2] shows explosive growth as beta increases	beta > 300, m18.out

Chain of Custody

Item	SHA-256
m18_resonance_ladder.py (script)	a311cb81809b786236a6ccc5d7c86a91751c6e84c421bb846e61...
m18.out (stdout)	93d6b554820ba699a522b9c68367928864d84de5fc8158880c64...
Parent M1 (alpha_0 = 299+pi/10)	63ef870a78766619327e99b68683bceff8c8ef9a525298756c77...
Parent M5 (C(S4) BC certified)	9df98a3970acbb6942770a6cdd42fb21b009f9a5f45a222dd963...
Parent M9 (g<=33 certified)	5e39f3a957d818fa85dad0a66d98a3c51801ba107ecea5a6bb45...

Conclusion

The resonance ladder sweep from k=0.50 to 3.50 (step 0.05, primes <=100000) confirms: (1) k=1.00 (beta_0) gives C=11.422, g_max=33 exactly matching M5/M9. (2) The explosion near k=3.18 (beta->300) is real: |S|=14, C=58.26, g_max=849. (3) k_c for c/10^6 = 2.5225 (not 2.67 as annotated in external chat). (4) External values at k=2.67 and k=3.18 differ from ours because the external computation used primes <= ~191 only; our search uses primes <= 100000. External values at k=2.00 and k=3.00 agree with ours, confirming the methodology is the same for small primes. All certified values SHA-bound in m18.out.

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